WHEN DOES REFERENCE MATCHING TRANSFER? SPECIFICATION-CERTIFIED AUDITS OF FIR/IIR REALIZATIONS

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ABSTRACT

Reference-based correctness scoring treats proximity to selected realizations as a proxy for satisfaction of a digital-filter specification, even when a magnitude mask admits many legitimate FIR or IIR designs. We audit that proxy on 20 magnitude-mask tasks (16 FIR, 4 IIR) whose 412 constructed valid implementations are continuously certified, using coefficient and magnitude-response catalogs of observed valid realizations. No observed valid coefficient reference recovers exact single-center membership on any task; exact coefficient catalogs require more than 10 references on all 20 tasks (median 23). After those catalogs were frozen, a catalog-blind protocol produced 614 new continuously certified valid realizations. The frozen coefficient catalogs accepted $66/614$ (10.7%; task-macro median 4.8%), whereas the frozen magnitude-response catalogs accepted $585/614$ (95.3%; task-macro median 100%). Reference adequacy therefore depends on representation and on the diversity of valid realizations that must be supported: coefficient matching is tied to realization identity, while magnitude-response matching is a substantially stronger—but not universally exact—surrogate on this suite. Code, frozen artifacts, and the reproduction entry point are at https://github.com/XianghuiMeng-1020/dsp-semantic-correctness.

***Index Terms*—** Digital filters, FIR, IIR, specification testing, reference matching

1. INTRODUCTION

A digital-filter task is specified by passband and stopband edges, ripple, attenuation, and, for IIR filters, stability [1, 2, 3, 4]: a feasible set, not a unique coefficient vector. Evaluation nevertheless often scores proximity to a golden coefficient vector, a reference magnitude response, or a small library catalog. That surrogate asks whether a candidate is the *same realization*, not whether it satisfies the mask. This paper audits whether such realization-reference oracles transfer to alternative specification-valid FIR/IIR designs.

That Remez, window, least-squares, frequency-sampling, Butterworth, Chebyshev, and elliptic constructions can meet one mask with different taps or sections is classical filter theory [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15] and is not our contribution. The general test-oracle problem, specification-based conformance, and prototype or set-cover selection are likewise established [16, 17, 18, 19, 20]. We do not propose a new oracle theory or a new set-cover algorithm. The question is whether coefficient or magnitude-response references that exactly fit one specification-certified FIR/IIR universe transfer to later catalog-blind implementations of the same masks.

This paper makes three contributions. (i) We formulate correctness as membership in a specification-defined valid set $\mathcal{V}_t$ and assemble a continuously certified FIR/IIR evaluation universe. (ii) We quantify how finite catalogs of *actual* valid realizations recover that membership on the frozen universe, including an exact observed-valid catalog-complexity diagnostic. (iii) On a catalog-blind prospective challenge of 614 independently certified realizations, we show a strong representation-dependent contrast: frozen coefficient catalogs transfer poorly, frozen magnitude-response catalogs usually transfer well, and admitting the new valids substantially expands every exact coefficient catalog.

2. SPECIFICATION SETS AND REFERENCE CATALOGS

A task t asks for an FIR impulse response or an IIR pair (b, a). A predicate $S_t(h) \in \{0, 1\}$ encodes the registered magnitude mask and, for IIR designs, a strict pole-radius bound. Correctness is membership

$$\mathcal{V}_t = \{h : S_t(h) = 1\}. \tag{1}$$

A reference catalog $\mathcal{R}$ is a finite set of *observed* specification-valid realizations. With a distance d and a common scalar threshold τ, the acceptance set is

$$\mathcal{A}_{\tau,\mathcal{R}} = \{h : d_{\mathcal{R}}(h) \le \tau\}, \quad d_{\mathcal{R}}(h) = \min_{r \in \mathcal{R}} d(h, r). \tag{2}$$

The reference oracle declares "correct" iff $h \in \mathcal{A}_{\tau,\mathcal{R}}$; the specification oracle declares "correct" iff $h \in \mathcal{V}_t$.

All exact-recovery statements below are restricted to an explicit finite universe $\mathcal{U}_t$. Write $V = \mathcal{V}_t \cap \mathcal{U}_t$ and $I = \mathcal{U}_t \setminus \mathcal{V}_t$, and define

$$D_V(\mathcal{R}) = \max_{v \in V} \min_{r \in \mathcal{R}} d(v, r), \quad D_I(\mathcal{R}) = \min_{i \in I} \min_{r \in \mathcal{R}} d(i, r). \tag{3}$$

Observation 1. On a frozen finite universe, a common scalar threshold recovers membership exactly if and only if $D_V(\mathcal{R}) < D_I(\mathcal{R})$. The statement is evaluation geometry, not a theorem over all filters.

The observed-valid *reference catalog complexity* $K^\star_{\text{obs}}$ is the smallest $|\mathcal{R}|$ of specification-valid occupants of V for which Observation 1 holds. Computing that finite-universe diagnostic reduces to a standard set-cover optimization [20, 19]; the optimization itself is not our contribution. $K^\star_{\text{obs}}$ measures realizable reference burden, not algorithmic novelty.

Two distances are used. Coefficient distance is a relative ℓ_2 after canonicalization (FIR: trailing-zero and sign rules; IIR: $a_0 = 1$, concatenated (b, a)). Magnitude-response distance is the RMSE of $|H|$ on the constrained pass and stop bands at $N_f = 131072$ frequency samples. Centers must be actual observed valid realizations, not arbitrary Euclidean points.

A length-31 Kaiser-window low-pass design and a length-45 weighted-Chebyshev (Parks–McClellan) design can both satisfy the same passband-ripple and stopband-attenuation mask while differing substantially in length, tap values, and phase-delay structure, so their coefficient vectors sit far apart under ℓ_2 even when their constrained-band magnitude responses nearly coincide. Coefficient distance is therefore sensitive to which valid realization was observed; magnitude-response distance is comparatively insensitive to that choice. Intuitively, $K^\star_{\text{obs}}$ grows when V is geometrically interleaved with I under the chosen distance: a single center under-covers V or over-covers into I whenever some valid realization is farther from every candidate center than some invalid realization is from its nearest center. Because coefficient distance uses each realization's own tap or section parameterization as the operative geometry, and different design families place their occupants in different regions of that space, a coefficient-space center separating one family's occupants from invalids need not separate another family's occupants from the same invalids—the source of the observed $K^\star > 10$ requirement. IIR tasks add a further partition: the strict pole-radius bound constrains each design family's (b, a) occupants to a family-specific region of coefficient space before any mask criterion is applied, so an IIR coefficient catalog inherits both the mask-driven and the stability-driven separation the FIR case does not.

3. CERTIFIED UNIVERSE AND PROSPECTIVE PROTOCOL

The base suite comprises 20 non-unique magnitude-mask tasks: FIR low-pass, high-pass, band-pass, and band-stop at $f_s = 8$ and $16\,\text{kHz}$, each loose and tight (16 FIR); and IIR low-pass and high-pass at $8\,\text{kHz}$ (4 IIR). The frozen constructed valid corpus contains 412 implementations (336 FIR, 76 IIR).

Final labels are not self-certified by the construction checker. For FIR, $P(x) = |H|^2$ with $x = \cos\omega$ is certified continuously over each constrained band by exact-rational Bernstein or Sturm sign reasoning. For IIR, we certify strict Schur stability (pole radius $<\ 0.999$) and the sign of $P_B(x) - CP_A(x)$ on every constrained band. All $412/412$ constructed valids received continuous specification certification; there were no valid$\to$invalid contradictions, and all frozen mechanism and boundary invalids certified invalid. Cosine endpoint handling is a numerical/rational certificate, not a machine-checked continuous proof. Implementation details live in the repository.

The eight prospective design families were realized with standard SciPy signal-processing routines [21]: `remez` (Parks–McClellan), `firls`, `firwin2` (frequency sampling), and `firwin` (Kaiser/Hann windowing) for FIR; `butter`, `cheby1`, `cheby2`, and `ellip` for IIR. No family was excluded a priori, and each family's outputs were certified by the same continuous procedure as the base catalogs before being counted toward the 614 frozen valids.

Three questions organize the audit. RQ1: when does realization-reference matching recover specification membership on this frozen universe? RQ2: how large a realizable catalog is required for exact recovery? RQ3: do exact base-fitted catalogs transfer to later implementations that satisfy the same masks?

We first froze the base catalogs and their max-safe thresholds (the largest τ that accepts no frozen base invalid). We then protocol-locked generation across all 20 tasks and eight standard families—Remez, FIRLS, frequency sampling, and windowed FIR; Butterworth, Chebyshev I, II, and elliptic IIR—scheduled 960 attempts, continuously certified occupants, removed exact coefficient duplicates, and froze 614 new nonduplicate valid realizations (500 FIR, 114 IIR; every task; at least 13 per task) without accessing catalog identities, reference distances, transfer scores, or ambient centers. Only then did we evaluate transfer. An earlier same-order probe set participated in catalog fitting and is not this holdout. The 614 occupants are prospectively generated, catalog-blind, specification-certified realizations, not all possible filters.

Table 1. Catalog-blind prospective transfer on 614 specification-certified realizations, by magnitude-mask task (L/T: loose/tight). Coeff. and Resp. are the frozen coefficient- and magnitude-response-catalog acceptance rates; N is the number of prospective valids for that task.

Task	N	Coeff.	Resp.	Task	N	Coeff.	Resp.
FIR LP, L, 8k	44	6.8%	100%	FIR BS, L, 8k	40	32.5%	97.5%
FIR LP, L, 16k	44	4.5%	100%	FIR BS, L, 16k	40	5.0%	97.5%
FIR LP, T, 8k	26	3.8%	100%	FIR BS, T, 8k	13	0.0%	100%
FIR LP, T, 16k	26	0.0%	100%	FIR BS, T, 16k	13	0.0%	100%
FIR HP, L, 8k	41	19.5%	97.6%	IIR LP, L, 8k	41	17.1%	78.0%
FIR HP, L, 16k	41	2.4%	97.6%	IIR LP, T, 8k	18	0.0%	100%
FIR HP, T, 8k	24	50.0%	91.7%	IIR HP, L, 8k	42	4.8%	100%
FIR HP, T, 16k	24	12.5%	87.5%	IIR HP, T, 8k	13	0.0%	100%
FIR BP, L, 8k	42	2.4%	100%				
FIR BP, L, 16k	42	4.8%	100%				
FIR BP, T, 8k	20	15.0%	70.0%				
FIR BP, T, 16k	20	30.0%	75.0%				

Table 2. Exact recovery on the frozen base universe. "Non-sep." is a finite-universe single-center failure. $K^\star$ is the minimum observed-valid catalog size.

Repr.	Can. non-sep.	Best obs.	med. $K^\star$	Lib. exact
Coeff.	$20/20$	$20/20$	23	$0/20$
Resp.	$19/20$	$18/20$	17.5	$0/20$

Table 3. Prospective transfer of frozen base catalogs to 614 catalog-blind certified-valid realizations.

Repr.	Accepted	Pooled	Macro med.	$\geq$95%	<75%
Coeff.	$66/614$	10.7%	4.8%	$0/20$	$20/20$
Resp.	$585/614$	95.3%	100%	$15/20$	$1/20$

4. RESULTS

4.1. Base-universe adequacy

On the frozen 412-valid universe, no single observed valid coefficient reference recovers exact membership (20/20 canonical and 20/20 best-observed non-separable). A single magnitude-response reference fails on $19/20$ canonical and $18/20$ best-observed tasks (Table 2). Exact observed-valid coefficient catalogs exist on $20/20$ tasks, but $K^\star > 10$ on every task (median 23). Library-only catalogs recover membership on $0/20$ tasks in both representations. Response $K^\star$ has median 17.5 and equals 1 on two tasks.

An arbitrary Euclidean center separates 19/20 coefficient and 20/20 response base universes (one IIR low-pass task has no coefficient center), so the observed failures should not be read as intrinsic one-center inseparability; the restriction to actual realizations is consequential.

4.2. Prospective transfer

This is the primary empirical result (Table 1, Table 3). Frozen coefficient $K^\star_{\text{obs}}$ catalogs accepted $66/614$ prospective valids (pooled 10.7%; task-macro mean 10.6%, median 4.8%; range 0–50%). Every task transferred below 75% (20/20); none reached 95%. FIR and IIR task-macro means were 11.8% and 5.5%. The same verdict is obtained under the predeclared midpoint threshold, so the coefficient failure is not an artifact of max-safe τ.

Frozen magnitude-response catalogs accepted $585/614$ (pooled 95.3%; task-macro mean 94.6%, median 100%). Fifteen tasks reached at least 95%; one tight FIR band-pass task transferred at 70%. FIR and IIR macros were both about 94.6%. Response transfer is generally strong and still task-specific.

Coefficient transfer failed across all eight generator families (family rates 3.6%–18.2%); response transfer for those families ranged from 85.7% to 100%. Coefficient transfer reached 0% on exactly three tasks—both tight FIR band-stop tasks and the tight FIR low-pass task at 16 kHz—where the certified prospective valids sit far, in tap space, from every frozen base-catalog member. Response transfer was weakest on the two tight FIR band-pass tasks (70% at 8 kHz, the only task below 75%; 75% at 16 kHz), followed by the tight FIR high-pass tasks (87.5%–91.7%) and the loose IIR low-pass task (78.0%).

Enlarging the coefficient catalog on the *base* universe does not rescue prospective transfer. Canonical $K{=}1$, published $K{\in}\{3,5\}$, and all-library coefficient oracles accepted $0/614$; the best observed single coefficient reference accepted 7.5%; the exact $K^\star_{\text{obs}}$ catalog accepted 10.7%. The corresponding response figures are 38.9% (canonical) and 95.3% ($K^\star_{\text{obs}}$). Increasing coefficient-catalog size enough to fit the original finite universe therefore does not produce strong prospective coefficient transfer.

4.3. Catalog maintenance

Admitting the 614 prospective valids and recomputing exact coefficient catalogs raises the median $K^\star$ from 23 to 55 (median relative growth 52.5%). Every task has $M^\star > 0$: every minimum-size expanded coefficient catalog requires at least one newly admitted realization (median $M^\star = 15.5$, maximum 37). In-sample catalog exactness therefore does not eliminate later support-expansion burden. Response catalogs grow much less (median $\Delta K = 0$; $M^\star > 0$ on $9/20$ tasks).

5. DISCUSSION

Coefficient matching retains realization-specific parameterization: distinct Remez, window, or elliptic occupants of the same mask can sit far apart in tap or (b, a) space, so a catalog exact on one sample can miss later valid designs $(66/614)$. Magnitude-response matching quotients out much of that coefficient-level variability and aligns more directly with the pass/stop mask S_t, which is why the same catalogs accept $585/614$ occupants. The contrast is a signal-processing representation fact, not merely that one rate is larger. Response matching remains a surrogate: it is not 100%, one task transferred below 75%, and it does not prove mask satisfaction.

A golden coefficient vector, or a finite coefficient catalog fitted to one implementation corpus, should not be read as a correctness oracle for a non-unique filter-design task without an adequacy audit. Where magnitude-response matching transfers, it can be a useful surrogate. The criterion used to define correctness in this study is still S_t. For evaluation practice, a reference-based filter-design checker should record which representation its catalog uses and should periodically recompute $K^\star$-style adequacy diagnostics as new valid realizations are observed, rather than treating one-time exactness on a fixed corpus as a permanent guarantee: no coefficient catalog earns that guarantee here, and even a favorable magnitude-response catalog needs periodic reappraisal.

This audit targets magnitude-mask FIR/IIR filters realized as direct-form coefficient vectors; cascaded, lattice, or multirate realizations [22] were not part of the certified universe and are left to future evaluation-methodology work. The study is finite: 20 magnitude-mask tasks, not general DSP software, and eight standard designer families, not all implementations. A secondary catalog-blind invalid challenge, covering $12/20$ tasks (310 realizations) and therefore not a balanced 20-task holdout, found response false acceptance ($96/310$, 31.0%) exceeding coefficient false acceptance ($38/310$, 12.3%) on that subset; full per-task results are in the repository. No infinite-universe claim is made, and magnitude-response matching is not asserted to be sound or complete in general.

6. CONCLUSION

Reference matching should be audited as a surrogate for a digital-filter specification, not assumed equivalent to it. Across 614 prospectively generated valid FIR/IIR realizations, coefficient catalogs that were exact on one implementation universe transferred poorly, while magnitude-response catalogs were substantially more stable. Correctness evaluation should distinguish specification compliance from realization identity and choose the reference representation accordingly.

Returning to the three questions: RQ1 (does realization-reference matching recover specification membership on the frozen universe?) is answered negatively for a single coefficient reference and only partially for a single response reference; RQ2 (how large a realizable catalog is required?) needs more than ten coefficient references on every task, with a median of 23; RQ3 (does an exact base-fitted catalog transfer?) is answered asymmetrically, with coefficient catalogs generalizing poorly $(66/614)$ and magnitude-response catalogs generalizing well but imperfectly $(585/614)$. The same audit methodology—freeze catalogs, generate a catalog-blind challenge, and measure transfer by representation— could be applied to other specification types, such as phase-response or group-delay masks, without requiring a new correctness definition.

Code and frozen artifacts are in the accompanying public repository, where the certified base and prospective corpora, the $K^\star$/RCC diagnostics, and a one-line reproduction entry point are frozen and tagged so that every number in Section 4 can be regenerated independently.

Compliance with ethical standards. No funding was received for conducting this study. The authors have no relevant financial or nonfinancial interests to disclose. The study did not involve human participants, animal subjects, or personally identifiable data.

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